

WEAPONISED BLUETOOTH TRACKING OF POLICE

Technical Analysis of a Real-World Proof-of-Concept Weaponisation

CONFIDENTIAL — RESPONSIBLE DISCLOSURE

Distributed BLE-Based Detection of Axon Law Enforcement Equipment

OUI 00:25:DF — Taser International / Axon Enterprise

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Date of Issue

28 April 2026

Classification

CONFIDENTIAL — Restricted to authorised disclosure recipients

*This document contains a technical description of a weaponisable system identified during ethical analysis.
It is intended solely for relevant law enforcement, regulatory and vendor recipients under responsible disclosure.
Unauthorised distribution, replication, or operational use of any system described herein is strictly prohibited.*

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1. Executive Summary

This report documents a working proof-of-concept system, recovered for the purpose of responsible disclosure, that turns cheap, off-the-shelf Bluetooth Low Energy (BLE) hardware into a **distributed mesh of independent radio scanners** capable of detecting, tracking and displaying the real-time location of law enforcement personnel.

The system has three pieces, and that is all. Many small low-cost scanner units, each operating completely independently, each listening only for the radio signature emitted by **Axon Enterprise** (formerly Taser International) law-enforcement equipment — body-worn cameras, smart holsters, conducted-energy weapons (Tasers) and in-car video systems. Whenever a scanner hears one of these signatures it sends a short message to a single central cloud server. The server fans every such message out, in real time, to a web dashboard that displays every detection on a live map as a moving notification feed for an operator. Each scanner knows nothing about any other; the cloud server stitches their independent reports into a city-wide picture.

The targeted radio signature is the manufacturer-assigned MAC-address prefix **00:25:DF**, which the IEEE has registered to Taser International / Axon Enterprise. Every Axon BLE-emitting product worldwide carries this prefix in the open. The system has no need to break encryption, no need to compromise police networks, and no need to pair with anything; it simply listens for a manufacturer code that Axon equipment is required to broadcast in order to function.

Each scanner unit can be assembled for under **USD/AUD 100**. It consists of a 4G LTE-connected BLE scanner, a small lithium battery, a low-wattage solar panel and a sealed waterproof enclosure. Once assembled, the unit can be discreetly placed on a rooftop, in a roof gutter, on top of an advertising sign, inside a utility cabinet, in a parked vehicle or at any other elevated or concealed location and left to operate indefinitely with no maintenance. With clear line-of-sight, an amplified RF front-end and a directional antenna, a single scanner can detect Axon equipment at distances of **up to approximately 400 metres**.

This document describes the architecture of that mesh, the components, the implementation choices, the deployment characteristics and the operational consequences. It is intended for relevant law enforcement agencies, parliamentary representatives, national-security bodies and regulators under standard responsible-disclosure principles.

A note on simplicity. The system was designed and written by hand, using ordinary open-source libraries; **no AI coding assistant or AI model was used in its construction**. The Python scanner is approximately 240 lines of straightforward procedural code. The cloud backend is a small set of standard ASP.NET Core controllers. The dashboard is a single React page. There is no machine learning, no novel cryptography, no custom radio firmware, no exploit of any vendor product, and no clever attack. The threat is precisely that the system is **so simple**: it is well within the capability of any competent hobbyist software engineer, and the defence against it cannot be a software patch — it must be a change in how Axon equipment broadcasts on the air.

Bottom line: a single attacker, with under USD 100 of hardware per node, a commodity cloud account and roughly 600 lines of Python and C#, can stand up an independent mesh of scanners that produces a real-time city-wide police-tracking dashboard — capability that would historically have required the resources of a state intelligence service.

2. Background — The Underlying Vulnerability

Every Bluetooth device has a unique Media Access Control (MAC) address. The first three octets of that address — the OUI — are assigned by the IEEE to a specific manufacturer. The OUI **00:25:DF** is registered to *Taser International, Inc.*, which now trades as **Axon Enterprise, Inc.**

Modern Axon equipment uses BLE for legitimate operational features. Examples include:

- **Signal Sidearm:** a holster sensor that broadcasts when an officer's firearm is drawn, automatically activating nearby Axon body cameras.
- **Body-worn camera pairing:** BLE used to pair Axon Body 2 / 3 / 4 cameras with phones, controllers and Axon Aware infrastructure.
- **Taser 7 / Taser 10 device telemetry:** conducted-energy weapons broadcast status and trigger events over BLE.
- **Axon Fleet in-car video:** dash-mounted units that pair with body cameras over BLE.

To perform any of these functions, the equipment must continuously broadcast BLE advertising packets containing its MAC address. The Bluetooth specification provides for *Resolvable Private Addresses* (RPA) — rotating, privacy-preserving identifiers that would prevent passive long-term tracking. Axon equipment, however, broadcasts using its **fixed, public OUI-prefixed MAC address**.

Layperson explanation. Police body cameras, Tasers and other electronic kit made by the company Axon all silently shout an electronic serial number into the air using Bluetooth, dozens of times per second, to a range of anywhere from a few metres up to roughly 400 metres in clear conditions with the right antenna. The first part of that serial number is the same on every Axon device in the world — it is the manufacturer's electronic 'name tag'. Anyone with a phone or a small embedded computer can quietly listen for that signature and, without any password or special permissions, know that an officer wearing Axon gear is nearby. The vulnerability is not a software bug; it is the basic decision to broadcast a hard-coded, manufacturer-tied identifier in the clear.

This issue was first raised publicly at **DEF CON 31 (2023)** in talks by Null Agent and "Sally, who makes yachts". A simple proof-of-concept Python script presented at and after those talks demonstrated single-laptop reception of Axon BLE advertisements.

Prior disclosure to Victoria Police (late 2022). The underlying detectability of Axon law-enforcement equipment via its broadcast OUI was first formally disclosed to **Victoria Police in late 2022** by the author of this report, predating the public DEF CON 31 talks. That earlier disclosure described the radio-frequency signature, the manufacturer attribution and the basic single-listener detection method. The present report is a follow-on disclosure, prompted by the recovery of a fully productionised, city-scale weaponisation of the same vulnerability — i.e. evidence that the threat has moved well beyond a research curiosity and into a deployable operational capability.

What is novel and serious about the system in this folder is not the discovery of the OUI signature — that has been known and previously disclosed — it is the productionised **weaponisation**: turning a single-laptop demonstration into a polished, distributed mesh of independent scanner nodes that share their findings through a single cloud-hosted dashboard. The result can be replicated and scaled to cover entire cities and beyond.

3. System Architecture — A Mesh of Listening Posts

The system is conceptually very simple. There are three pieces — a fleet of scanners in the field, one cloud server that receives their reports, and one web dashboard that displays the result to an operator — and the whole point of the design is to make adding more scanners trivial.

Global explanation, in one paragraph: Imagine fifty (or five hundred, or five thousand) small sealed boxes scattered across a city — on rooftops, inside utility cabinets, in parked cars, taped under benches. Each box runs the same little program. Each box listens on its own, knows nothing about any other box, and is the size of a pack of cards. When any box hears a Bluetooth 'hello' from an Axon

device, it sends a single short message over its 4G connection to one website. That website re-broadcasts the message, within a second, to anyone with the dashboard open in a web browser. The dashboard plays a sound, drops a glowing dot on a live map, and adds the sighting to a notification feed. The system is therefore a **passive sensor mesh feeding a live notification dashboard** — exactly like an air-traffic-control screen, except the 'aircraft' are individual police officers and patrol cars.

Layer	Component	Technology	Function
1. Scanner mesh	Many independent BLE scanner nodes	Python 3 + Bleak on a low-cost Linux SBC; each node has 4G, GPS, battery and solar.	Each node passively listens for BLE advertisements with MAC OUI 00:25:DF. On every detection, the node POSTs a small JSON message — MAC, derived serial, GPS, signal strength — to the cloud server. Every five seconds it also sends a heartbeat. Nodes do not communicate with each other.
2. Cloud server	Single central API + real-time hub	ASP.NET Core 6 + SignalR + in-memory cache	Receives detection messages from any number of scanners, stores them, and instantly fans them out as WebSocket events to every connected dashboard client. Acts as the only shared point of the system.
3. Web dashboard	Real-time notification UI	React + Leaflet, in a web browser	Displays every detection as a live map marker and as a scrolling notification feed. Plays an audio cue on each new detection. Builds a movement trail per Axon device by stitching together the reports from independent scanners. This is what the operator looks at.

Diagrammatically, the data flow is: **Axon device** → **independent scanner** → **cloud server** → **web dashboard** → **operator**. Each step adds zero round-trip latency that the operator can perceive: from the moment a Bluetooth advertisement is received in the field to the moment a marker appears on the operator's screen is typically under one second. There is no peer-to-peer co-ordination between scanners; the cloud server is the only place where the picture becomes global.

4. Component-Level Analysis

The following sections describe each component as found in the recovered source tree.

4.1 BLE Scanner Node — the field-side listening post

A single Python file (~240 lines) implementing the entire field-side sensor. Targeting and behaviour are controlled by a small block of constants which a deployer edits once before flashing the SD card. The constants identify the manufacturer OUI to detect (00:25:DF), the address of the cloud backend, and a unique scanner identifier so that detections from this node can be distinguished from detections by every other node in the mesh.

The scanner uses the cross-platform *Bleak* library to subscribe to every BLE advertising packet in range. On every advertisement, the callback simply checks whether the device MAC begins with the Axon OUI. If it does, and the MAC has not been seen during this session, an alert is built and POSTed to the cloud:

```
if mac_address.startswith(MAC_PREFIX):
    if mac_address not in self.detected_devices:
        ...
        self.send_alert(device)
```

The alert payload includes the full target MAC, a derived 6-character serial (the last six hex digits of the MAC), the GPS coordinates of the scanner at the time of detection, the received signal strength (RSSI), and a small rolling history of RSSI readings. The RSSI history allows the backend to estimate distance and to construct a movement / signal-strength trail per device. A heartbeat is sent every five seconds so the cloud knows which scanner nodes are currently live.

Reception range. A standard BLE radio receives advertising packets line-of-sight at approximately 10–30 m for a Class-2 transmitter and up to ~100 m for a Class-1 transmitter. Practical maximum reception against Axon equipment using an amplified RF front-end and a directional antenna has been demonstrated at **up to approximately 400 m**. A single node with that hardware therefore covers an area of up to ~50 hectares (a circle of radius 400 m).

Operational characteristics. The node is purely passive: it makes no transmissions to police equipment, performs no decryption, and does not pair with any device. It uses only lawful radio reception. It is cheap: total bill of materials per node, including a 4G LTE modem, GPS, BLE radio, low-wattage solar panel, lithium battery, and a small waterproof IP-67 enclosure, is under **USD/AUD 100**. Once sealed, the unit can be discreetly placed on a rooftop, in a gutter, on top of an advertising sign, inside a utility cabinet, in a parked vehicle, or at any other elevated or concealed location and left to operate indefinitely. It is stealthy: it produces no externally visible RF emissions beyond ordinary outbound 4G cellular data traffic, which is indistinguishable on-air from a phone or a tablet.

4.2 Cloud Backend — the single point that joins the mesh

Hosted on a commodity public cloud platform. The backend exposes a small set of HTTP endpoints and a real-time WebSocket hub for event distribution. The principal controllers are:

- **AlertController** — exposes endpoints for receiving detections, scanner heartbeats, and scanner / alert / statistics queries. Validates incoming detections, stores them in an in-memory cache (max 1000 alerts, 24-hour sliding window), and immediately broadcasts every detection to all connected dashboard clients via the real-time hub.
- **AlertHub** — a SignalR WebSocket hub that pushes *ReceiveAlert* and *ScannerUpdate* events to every subscribed client in real time.

Notable design choices that shape the threat:

- The recovered configuration accepts requests from **any origin** with credentials, so any web page on any domain can establish a session and stream live detections.
- There is **no authentication or authorisation** on any endpoint. Anyone who knows the address can submit detections, read live data or scrape historical alerts.
- Alerts persist in an in-memory cache for 24 hours with a 1000-event ring buffer; longer storage in a managed database is on the published roadmap.

4.3 Web Dashboard — the operator's notification screen

A single-page web application built with React 18 and Leaflet.js. It connects to the cloud backend over a SignalR WebSocket and renders detections in real time. Major features:

- Live **Leaflet** map (street, satellite, dark or light tiles) showing every BLE scanner node and every detected Axon device, with pulsing markers for fresh detections.
- Distinct icons for new vs. observed devices and for scanner online/offline state.
- Tabs for a live map, a raw-data terminal feed, an interaction log, and a device grid view.
- Audio alerting on every new detection; auto-centring of the map on the most recent detection.

- Per-device path drawing: stores up to 50 historical positions per MAC in the browser and renders the resulting movement trail on the map.
- Export to CSV and JSON of all stored data.

An obfuscation pipeline is included alongside the source, indicating the operator's intent to deploy the dashboard in a form that resists casual reverse engineering.

5. How the Weaponisation Works

Considered in isolation, each component is benign or borderline. Considered together, they form a purpose-built law-enforcement evasion system. The following describes how a single deployment operates from the perspective of an attacker.

5.1 Technical operation

- **Step 1 — Build the scanner mesh.** The attacker assembles N units. Each unit is a small single-board computer with onboard or USB BLE, a 4G LTE modem with prepaid SIM, a GPS receiver, a lithium battery, a small solar panel, and a sealed waterproof enclosure. Bill of materials per node is under USD/AUD 100.
- **Step 2 — Deploy.** Nodes are placed in concealed elevated or weather-protected locations — rooftops, gutters, advertising-sign cavities, utility cabinets, parked vehicles, balconies — along corridors of operational interest. A sealed solar-powered unit on a rooftop can run indefinitely with no maintenance. Each node operates entirely independently.
- **Step 3 — Continuous passive scanning.** Each node runs the scanner script as a system service. It scans constantly, fingerprints any device whose MAC begins with 00:25:DF, and POSTs an alert to the cloud backend over its 4G link.
- **Step 4 — Cloud aggregation.** The backend receives the alert, stores it, and rebroadcasts it via SignalR to every dashboard client in approximately 100 milliseconds.
- **Step 5 — Dashboard notification.** The dashboard renders the alert as a glowing marker on a live map, plays an audio cue, and adds the device to a per-officer movement trail. The same MAC detected by multiple independent nodes produces a sequenced trail across the city — the mesh effectively triangulates a moving officer without any node knowing it is doing so.
- **Step 6 — Operator action.** An operator viewing the dashboard, in real time, knows the current position of every nearby Axon-equipped officer and can route around them.

5.2 Laverson explanation

Imagine this. A criminal organisation wants to know — at all times, in real time, with a map on a phone — exactly where every police officer in a city is currently standing or driving. They cannot hack police radios; that is illegal and difficult. They do not need to. Police body cameras and Tasers, made by a US company called Axon, broadcast a small electronic 'hello' over Bluetooth roughly thirty times per second. That 'hello' contains a manufacturer code that uniquely says *this is Axon kit*.

The criminal places a small sealed box — costing under USD 100 — at fifty locations across a city: on rooftops, inside power-pole junction boxes, taped under benches, inside parked cars. Each box has a Bluetooth radio, a 4G SIM card, a small solar panel, a battery, and nothing else. Each box works alone. None of them talks to any of the others. Each is just listening for the Axon 'hello'. When it hears one, it sends a short message over the mobile network to a single website, saying *Axon device 9F-3C-2A is at this exact location, signal strength -57*. The website immediately pushes that message to a dashboard, open in a web browser on a phone in a fast car, which displays a live map: a glowing red

dot anywhere a police officer is currently standing. The whole city lights up like an air-traffic-control screen, except the 'aircraft' are individual officers and patrol cars.

Neither the police officer wearing the camera, nor Axon, nor anyone else has any way to detect that this is happening. There is no malware on police equipment. There is no breach of police networks. The system uses only public radio waves that the equipment is required to broadcast in order to function.

6. Scaling the Mesh Across Large Geographic Areas

This section addresses the most operationally significant property of the recovered system: the mesh is engineered from the ground up for **horizontal scale**. There is no single-point-of-failure that limits the geographic footprint, and the marginal cost of adding capacity is small. A deployment can be grown from a single street to a whole-of-state region using the same code, the same backend and the same dashboard — the only change is the number of physical scanner nodes in the field.

6.1 Architectural properties that enable scale

- **Stateless edge nodes.** Each scanner is a single Python process holding only a short-lived RSSI history in RAM. Nodes know nothing about each other. Adding a new node is exactly as expensive as deploying its hardware — there is no cluster to join, no key exchange and no orchestration.
- **Many-to-one ingestion.** The cloud accepts an unlimited number of independent scanner identifiers. The recovered code already partitions per-scanner state, so 1, 100 or 10 000 scanners look identical to the backend.
- **Cloud elasticity.** The backend runs on a managed cloud platform that auto-scales horizontally. The published roadmap explicitly contemplates promotion to a hosted real-time messaging service, lifting the per-instance WebSocket connection limit and enabling concurrent dashboard fan-out into the tens of thousands.
- **Persistent storage roadmap.** The current 1000-event in-memory cache is the prototype path; the published roadmap names a managed SQL database and a NoSQL document store as the production options. Both options give global, multi-region writes — a deployment could span a continent from a single tenant with no code changes.
- **Independent ownership of nodes.** Each scanner is a self-contained device. Seizure or compromise of one node does not degrade the remainder. Nodes can be rotated, replaced or moved arbitrarily.
- **Bounded telemetry per node.** A single scanner produces only a heartbeat every five seconds plus a small alert per detected device. Even a fleet of 1 000 nodes generates well under 1 MB/s of ingest traffic — within the free tier of most cloud providers.

6.2 Sensor density and effective coverage

BLE has a typical line-of-sight range of approximately 10–30 m for Class-2 transmitters and up to ~100 m for Class-1 transmitters. Coverage of a road is therefore roughly a 30–60 m disc around each scanner using stock hardware. With an amplified RF front-end and a directional antenna, demonstrated maximum range against Axon equipment is **up to approximately 400 m**, extending each node's coverage to as much as ~50 hectares. Approximate node counts required to provide effective coverage of common geographies, assuming stock BLE range:

Coverage target	Approx. road length	Density	Approx. node count
Single inner-city neighbourhood	~30 km of road	Continuous arterial coverage	60–80
Inner-metro central area	~300 km of road	Continuous arterial coverage	500–800
Whole metropolitan region	~9 000 km of road	Arterials only	2 000–4 000
Whole state — major roads only	~22 000 km of road	Highway + town centres	5 000–10 000
Whole country — capital cities only	~50 000 km of road	Capital arterials + highways	20 000–40 000

These figures assume static, fixed-position scanners with stock BLE range. They are conservative because they ignore the force-multiplying effects of mobile nodes (covered in §6.4) and because the use of amplified or directional antennas — which are inexpensive — can extend each node's effective radius from ~30 m to ~400 m, halving or quartering the required node counts in the table above.

6.3 Cost to scale

The economic profile is what makes this threat serious. A self-contained, weather-sealed, solar-powered, 4G LTE-connected BLE scanner can be assembled from off-the-shelf components for **under USD/AUD 100**. Indicative bill of materials per static node:

- Single-board computer with onboard or USB BLE radio: ~AUD 30
- USB or HAT-mounted GPS receiver: ~AUD 10
- 4G LTE modem hat + prepaid SIM: ~AUD 30 hardware + ~AUD 10/month data
- Lithium battery + small solar panel + charge controller: ~AUD 15
- Sealed IP-67 waterproof enclosure with antenna grommets: ~AUD 10
- **Total per static node: under AUD 100; ongoing: ~AUD 10–15/month for cellular data.**

Indicative total cost-of-ownership for representative deployment scales (capex + first 12 months of opex):

Scale	Nodes	Capex (AUD)	Year-1 Opex (AUD)	All-in Year 1 (AUD)
Neighbourhood	80	\$8k	\$10k	\$18k
Inner metro	600	\$60k	\$72k	\$132k
Whole metro	3 000	\$300k	\$360k	\$660k
State arterials	8 000	\$800k	\$960k	\$1.76M
Country capitals	30 000	\$3.0M	\$3.6M	\$6.6M

Critical observation: a deployment large enough to provide continuous, real-time, city-wide officer-position awareness costs less than the price of a single luxury vehicle. A state-wide deployment costs less than the price of a modest property purchase. The historic barrier to this kind of distributed surveillance was capital expenditure on RF infrastructure; that barrier no longer exists.

6.4 Force multipliers — mobile nodes

Static scanner counts overstate the requirement, because the same code base trivially supports mobile and opportunistic deployments:

- **Vehicle-mounted nodes.** A scanner running in a moving vehicle covers an effective swathe equal to its BLE range × travelled distance. A single courier or rideshare vehicle driving 100 km of arterials per

day passes every BLE-emitting Axon device within range of its route. Twenty such vehicles substitute for several hundred static nodes.

- **Smartphone nodes.** Bleak runs on Android via Termux, and equivalent BLE scanning APIs exist on iOS. A mobile app distributed to a network of operators turns each phone into a sensor — converting a recruited human network into a passive surveillance grid at zero hardware cost.
- **Rideshare / courier piggybacking.** Devices physically attached to rideshare vehicles, food-delivery scooters, parcel vans and taxis can blanket a city with no additional driver.

6.5 Cross-jurisdictional scaling

Two properties of the recovered architecture make cross-jurisdictional, multi-state or international scaling effectively trivial:

- **The OUI is global.** Axon equipment ships to police agencies in many countries. A scanner deployed anywhere in the world will detect any Axon device anywhere in the world, with no localisation. The same software stack works in any jurisdiction without modification.
- **Multi-tenant by design.** The published "Future Enhancements" list in the recovered documentation explicitly names *Multi-tenancy support* and *Geofencing with custom zones* — i.e. the operators themselves contemplated franchising the system to multiple cities or organisations from a single backend.

6.6 Operational scaling pattern

A realistic, end-to-end scaling sequence that an organised group could plausibly execute over 12 months, starting from the recovered code base:

- **Month 0–1.** Run a single scanner on a laptop in a single area. Validate that detections arrive and the dashboard works. Total cost: under AUD 200.
- **Month 1–3.** Build 50 static nodes and seed them along the arterials of one inner-metro region. Recruit 10 mobile operators. Total cost: ~AUD 5 000.
- **Month 3–6.** Expand to 500 nodes covering whole-of-metro arterials. Migrate the in-memory backend to a persistent database for long-term intelligence retention. Total cost: ~AUD 50 000.
- **Month 6–9.** Roll out vehicle-mounted nodes through a sympathetic delivery or rideshare network. Coverage expands to all arterials and most residential streets.
- **Month 9–12.** Deploy a sister tenant in a second city. Same backend, different scanner namespace. The system is now national.

Key take-away: the work has already been done. The recovered source tree is not a starting point — it is a finished product. Scaling it from a single demonstration to city-wide, then to state-wide, then to national coverage requires only money for hardware and SIM cards, plus people willing to deploy boxes. None of these steps requires further software development, research or specialist skills.

7. Potential Danger

The danger this system represents falls into operational, officer-safety, public-safety, intelligence and vendor-trust categories.

7.1 Technical danger

- **Loss of officer locational privacy.** Any uniformed officer carrying Axon equipment can be identified, tracked and historically logged by an unaccountable third party.

- **Real-time evasion of policing.** Couriers carrying contraband, organised-crime drivers, persons subject to warrants and persons evading apprehension can be routed around live officers with sub-second latency.
- **Targeted ambush enablement.** The same telemetry that allows evasion equally allows interception. An attacker can wait for a single-crewed patrol vehicle to enter a chosen location and mount a deliberate attack.
- **Force-composition disclosure.** Each Axon device has its own MAC. Aggregating MAC counts at a location reveals how many officers are present, and re-detecting the same MAC over time reveals an individual officer's roster, beat and routine.
- **Long-term pattern-of-life analysis.** Persistent storage of timestamped sightings produces a profile of every officer: home area, regular start/finish times, partner officers (frequently co-detected MACs), and habitual stops.
- **Counter-surveillance immunity.** Conventional bug-sweeps look for transmitters. These nodes do not transmit anything detectable on-scene; they only listen, then exfiltrate over ordinary cellular data. Existing counter-surveillance procedures are blind to them.
- **Replicability.** The entire stack is built from open-source libraries. Anyone with an undergraduate-level software-engineering ability can rebuild it from this report's description in a weekend.

7.2 Laverson danger

The most direct way to put this is: **a system like this is the difference between a single getaway driver guessing where police are, and an organised group watching every officer in the city on a moving map.** The same technology that lets an alarm system detect movement is being used to detect officers wearing the cameras, holsters and stun-guns made by Axon.

For everyday people, this matters in three ways. First, organised criminals using such a system can complete drug runs, robberies, weapons movements and assaults with a real-time guarantee that no officer is within several hundred metres of them. Second, anyone choosing to harm an officer can use it to choose an isolated single-crewed unit and arrange to be in the right place at the right time. Third, because the system also identifies officers as individuals (each piece of Axon equipment has a unique number), the same data can be used to harass, intimidate or threaten specific officers off-duty: their daily patterns and family routines could in principle be reconstructed from the data.

The hardest fact to convey to non-technical audiences is this: **the police themselves cannot see this happening.** Their radios are not compromised. Their networks are not breached. Their devices behave exactly as designed. The exposure comes from the ordinary, by-design operation of standard-issue Axon equipment, broadcasting a manufacturer code that distinguishes it from any other Bluetooth device on earth. Until that broadcast behaviour changes, every officer on duty is a moving beacon to anyone who chooses to listen.

8. Indicators That a Deployment is Live

If a deployment of this or a substantially similar system is operating in a given area, the following observable indicators may be present and could be used as detection criteria for a counter-deployment effort:

- Unaccountable small sealed enclosures (battery boxes, junction boxes, cavity panels) on rooftops, in gutters, on advertising signs, at the perimeter of police facilities, and along main roads.
- Concealed devices radiating only outbound 4G/LTE traffic — the scanner does not need to transmit on Bluetooth, only listen.

- Marked patterns of suspect vehicles consistently avoiding officers' physical positions in real time, despite no obvious means of awareness.
- Anomalous outbound data traffic from edge devices located in unusual physical locations.

9. Recommendations

9.1 For Axon Enterprise (the equipment vendor)

- Adopt **Resolvable Private Addresses (RPA)** on all BLE-capable products as the default, so that the broadcast MAC rotates and cannot be used as a long-term manufacturer-tied identifier.
- Where a stable identifier is required for legitimate operational pairing (e.g. holster ↔ camera), pair it cryptographically rather than broadcasting it in cleartext advertising packets.
- Reduce or randomise advertising intervals; consider reduced TX-power modes when the equipment is not in active use.
- Issue a security advisory to all customer agencies acknowledging the OUI-based discoverability of fielded equipment so that operational risk can be assessed.
- Provide a firmware path to disable BLE advertising for officers in covert or undercover roles.

9.2 For law enforcement agencies

- Conduct a risk assessment of the fielded fleet of Axon and equivalent BLE-emitting equipment for surveillance exposure.
- Update counter-surveillance procedures to include passive BLE listeners as a recognised threat category. Consider periodic over-the-air sweeps for unauthorised 00:25:DF activity at known-clean locations.
- Treat the existence of unauthorised cloud endpoints aggregating 00:25:DF detections as an active investigative lead. Approach the relevant cloud platform operator with a lawful preservation request.
- Consider issuing officers with shielded equipment carriers when off-duty, or BLE-disabling options when not actively recording.
- Review contractual clauses with Axon and other vendors to require RPA-based identification.

9.3 For regulatory and oversight bodies

- Treat hard-coded broadcast manufacturer identifiers on safety-critical equipment as a category of **privacy-by-design failure** requiring industry guidance.
- Engage with the IEEE and the Bluetooth SIG to encourage RPA-mandatory profiles for law-enforcement and emergency-services use cases.
- Consider whether operating a system substantially similar to that described constitutes an offence under existing surveillance-devices and stalking legislation; explicit guidance would aid prosecution.

10. Responsible Disclosure

This report was produced by Tester Present Specialist Automotive Solutions and Benjamin Jack Leighton in good faith, on the basis that the analysed proof-of-concept materials came into our possession in the course of legitimate research. The materials have not been published, distributed or operated by us. No detection has been performed against any live Axon device or any serving officer.

Prior disclosure history. The underlying vulnerability — namely, that Axon law-enforcement equipment is uniquely identifiable in the air by its IEEE OUI 00:25:DF and can therefore be detected at range by any commodity Bluetooth receiver — was first formally disclosed to **Victoria Police in late 2022** by the author of this report. That earlier disclosure pre-dated the public DEF CON 31 (2023) talks on the same subject. The present report is therefore not a first disclosure of the vulnerability; it is a follow-on disclosure of the **operational weaponisation** of that vulnerability, made necessary by the recovery of the source tree analysed above.

The intended recipients of this disclosure are:

- **The relevant State Police service** — as the agency most directly exposed by the bounded deployment indicated in the recovered material.
- **The Australian Federal Police (AFP)** — for federal-level coordination.
- **The Australian Security Intelligence Organisation (ASIO)** — for assessment of the national-security implications of a domestically deployable, passively operated officer-tracking mesh.
- **The National Security Hotline of Australia** — as the standing intake channel for matters of national-security concern reported by members of the public.
- **The Australian Cyber Security Centre (ACSC / ASD)** — for technical coordination.
- **Members of the Parliament of Australia** — for legislative awareness, given the implications for officer safety, public safety and surveillance-devices law.
- **The Microsoft Security Response Centre and equivalent cloud-platform abuse channels** — to consider takedown and preservation of the identified hosting tenant.
- **The Bluetooth SIG and the IEEE Registration Authority** — for awareness of the broader profile.

We are willing to:

- Provide a sealed copy of the analysed source tree to authorised investigators;
- Brief technical staff at any of the recipients above in person or over secure video;
- Co-operate with any lawful investigation arising from the existence or operation of any live deployment.

This report is provided strictly on a CONFIDENTIAL, responsible-disclosure basis. It must not be distributed beyond the authorised recipients and their immediate technical and legal advisors. Any republication, even in part, must obtain the written consent of the author.

11. Glossary

- **BLE** — Bluetooth Low Energy. Short-range wireless protocol; advertises itself many times per second.
- **OUI** — Organisationally Unique Identifier. The first 24 bits of a MAC address, IEEE-assigned to a manufacturer.
- **MAC address** — 48-bit hardware identifier of a network or radio device.
- **Mesh** — used in this report to mean a fleet of independent scanner nodes that do not talk to each other but each report into the same central server.
- **RSSI** — Received Signal Strength Indicator (in dBm). Used to estimate range from sensor to target.
- **RPA** — Resolvable Private Address. A privacy-preserving, rotating BLE address mode.

- **SignalR** — Microsoft real-time WebSocket framework used to push events from server to clients.
- **Axon Enterprise** — US public company; vendor of body cameras, Tasers and related law-enforcement equipment. OUI 00:25:DF.

End of Report.

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Issued 28 April 2026. CONFIDENTIAL — Responsible Disclosure.

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